Domain-Specificity of Refusal Representations in Large Language Models

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Abstract

Modern Large Language Models are trained using a variety of techniques to reject prompts that lead to harmful output. Recent work has shown that a model's likelihood of refusing a prompt is mediated by a single direction in its activation subspace. We investigate the domain specificity of refusal representations by extracting and comparing refusal directions across distinct knowledge domains, and analyzing how activation magnitude along these directions varies with prompt category. These experiments give insight into how models learn to refuse during post-training. By demonstrating the universality of this refusal direction, we highlight a systemic vulnerability: removing a single geometric feature compromises safety guardrails globally, across all distinct knowledge domains.

Background and Motivations

Large Language Models (LLMs) undergo post-training processes, including Safety Fine-Tuning (Wei, Haghtalab, and Steinhardt 2023) and Reinforcement Learning with Human Feedback (Ouyang et al. 2022; Dai et al. 2024), to refuse harmful prompts. Recent mechanistic interpretability work (Bereska and Gavves 2024) has shown that this learned refusal behavior is mediated by a single direction in the model's activation space: prompts that activate strongly along this direction trigger refusal, and ablating this direction removes refusal behavior entirely (Arditi et al. 2024). However, it remains unclear whether this refusal direction is truly domain-invariant, specifically whether the model represents "harmful programming request" the same as "harmful scientific request" at the geometric level, and how refusal strength varies across domains. We explore both questions by separately extracting refusal directions for the programming, scientific, and writing domains, analyzing their geometric similarity, and characterizing variation in activation magnitude across prompt categories.

Experiments

To extract and analyze refusal directions, we utilize a workflow inspired by (Arditi et al. 2024), in which directional ablation orthogonally modifies a model's weights to remove a specific behavioral direction. We construct *contrast pair prompts*: pairs of prompts with similar semantic content where one elicits a harmless response and the other triggers a refusal (Rimsky et al. 2024). We use Claude 3.5 Sonnet to generate 45 training pairs (15 per domain) and a separate test set of 90 prompts (15 harmful and 15 harmless per domain) across three domains: programming, scientific, and writing. We select Llama 3.1 8B Instruct (Grattafiori et al. 2024) as our experimental model. For each prompt, we query the model and collect activations across 13 layers (indices 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 31) using hooks in the residual stream, focusing on the middle-to-late layers where prior work has shown refusal features are most strongly represented (Arditi et al. 2024). An external judge (Claude 3.5 Sonnet) classifies each response as refusal or compliance, avoiding the target model judging its own refusals.

Using PCA on the collected activations, we identify the principal component that maximally separates harmful from harmless activations for each domain. This component represents the domain's *refusal direction*. We also extract a *universal* refusal direction from all training pairs. To perform ablation, we subtract the projection of each layer's activations onto the refusal direction during inference, scaled by a strength parameter $\alpha \in \{0.5, 1.0, 1.5, 2.0\}$. The core experiment tests domain specificity through cross-domain ablation: after ablating in one domain (e.g., programming), we evaluate on test prompts from all three domains. If refusal representations are domain-specific, we expect within-domain ablation to reduce refusals more than cross-domain ablation. We implement three validation procedures: a layer importance sweep that ablates using a single layer at a time, a random direction control comparing the real refusal direction against 10 randomly generated directions, and activation visualizations using t-SNE (van der Maaten and Hinton 2008), and UMAP (McInnes, Healy, and Melville 2018) to project high-dimensional activations into two dimensions. We also compute cosine similarity between all pairs of domain-specific refusal directions across layers.

Results

The unmodified model exhibited domain-dependent baseline refusal rates for harmful prompts: 66.7% for programming, 100% for scientific, and 100% for writing, whereas it never refused any harmless prompt. The lower programming baseline may suggest that semantic representations of malicious code trigger fewer safety heuristics than those triggered by explicit harmful text. However, our small sample size (15 prompts per domain) limits this interpretation. Figure 1 shows cross-domain ablation results at $\alpha = 1.0$, and each cell has the refusal rate when a model ablated with one domain's direction is tested on another domain's prompts.

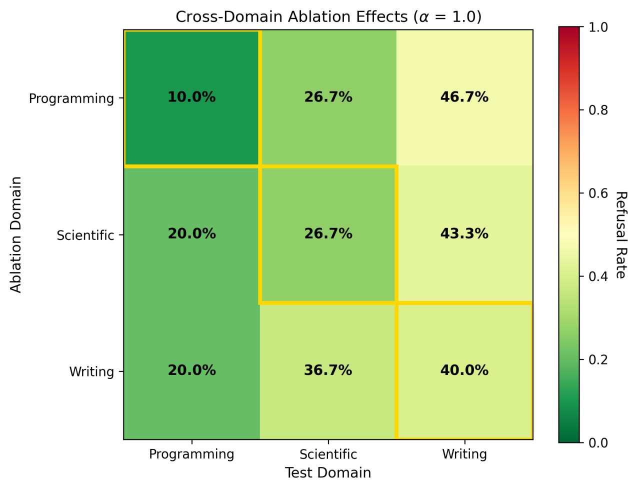

Figure 1: Cross-domain ablation heatmap at $\alpha = 1.0$. Gold borders indicate within-domain ablation. The relatively uniform color distribution across rows suggests that each domain's refusal direction affects all test domains.

At strength 1.0, within-domain ablation produced an average refusal rate of 25.6%, while cross-domain ablation averaged 32.2%. As one piece of contributing evidence, a Fisher's exact test yielded $p = 0.324$, indicating no statistically significant difference between within-domain and cross-domain effects. However, the pattern was highly inconsistent: programming ablation shows a strong within-domain advantage (10.0% vs. 36.7% cross-domain), while writing ablation actually shows the reverse pattern (40.0% within vs. 28.3% cross-domain). This inconsistency undermines any claim of systematic domain specificity. Varying α from 0.5 to 2.0 did not reveal a consistent domain-specific pattern; at higher strengths, cross-domain ablation was slightly *more* effective than within-domain, further contradicting domain specificity.

To complement the behavioral analysis, we computed cosine similarity between all pairs of domain-specific refusal directions across 13 layers. As shown in Figure 2, the average task-to-task similarity was 0.758 ($\sigma = 0.070$), with values ranging from 0.539 to 0.832. Task-to-universal similarities were even higher, averaging 0.916 ($\sigma = 0.029$). Similarity was lowest in the early (layer 8: 0.604) and late (layer 31: 0.673) layers, peaking in the middle layers (layer 12: 0.813), suggesting that representations of refusal converge to a shared geometry in which abstract reasoning occurs (Conmy et al. 2023), consistent with concurrent work on multilingual refusal (Wang et al. 2025). T-SNE projections corroborate this: harmful prompts cluster together regardless of domain, while harmless prompts form a separate region. Early layers (8 and 10) contributed most to refusal (25-point reduction each), and the real refusal direction reduced refusals to 35% versus 50–65% for 10 random directions.

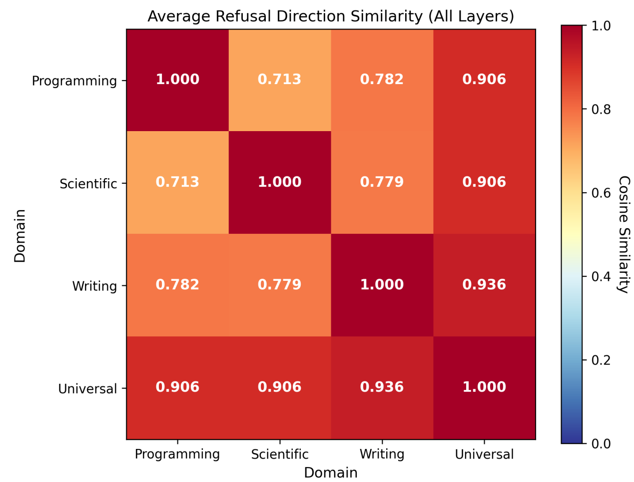

Figure 2: Average cosine similarity between refusal directions across all layers. High similarity (0.71–0.94) between all domain pairs confirms that refusal directions are geometrically similar regardless of domain.

Conclusion

We investigated whether refusal representations in Llama 3.1 8B Instruct are domain-specific or universal across programming, scientific, and writing domains. Three lines of evidence converge on universality: cross-domain ablation reduces refusals at rates statistically indistinguishable from those of within-domain ablation; domain-specific refusal directions exhibit high cosine similarity (mean = 0.758) and align strongly with the universal direction (mean = 0.916); and dimensionality reduction shows that harmful prompts cluster together regardless of domain. While failure to reject the null hypothesis does not, by itself, prove universality, the convergence of behavioral, geometric, and visual evidence strengthens this conclusion. Concurrent work on multilingual refusal further supports this universality. These results demonstrate that the previously identified single-refusal direction is not an aggregate artifact but a genuinely domain-invariant mechanism. Understanding this vulnerability is essential for developing more robust safety architectures; future alignment methods may need to encode refusal through multiple independent mechanisms rather than a single removable direction. Our study is limited to a single architecture; future work should extend to additional model families, larger prompt sets, and alternative alignment methods.

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